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& Biomedical Engineering**
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Programs: Electrical Engineering

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Instructor:	Dr. Kandasamy Illanko
Teaching Assistant (TA)	Kevin Lowell Pham - kevin.lowell.pham@torontomu.ca

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Student Last Name	Student First Name	Student ID	Section	Signature*
Abbas	Syed Maisam	501103255	05	

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INTRODUCTION

This lab assignment investigates the basic concepts of discrete-time signal processing to analyze mathematical models and digital implementations to calculate the unit impulse response, zero-input response, zero-state response, and total response for linear time-invariant discrete-time (LTID) systems. Previously, in Lab 1, recursion methods were used to determine the physical attributes (such as transformations, energy, and power) of discrete signals using algorithms and data structures in the Python programming language.

Throughout this lab, the following student number will be computed as per the given sequence of variables.

$$\begin{array}{l} \text{Student ID} = 501103255 \qquad [1] \\ [A = 5, B = 0, C = 1, D = 1, E = 0, F = 3, G = 2, H = 5, I = 5] \end{array}$$

This lab will identify the distinctive modes in order to solve the system equation. Additionally, the assessment of a system's stability will be computed iteratively by deriving equations using the given functions. The applications of how discrete systems respond to both input/output circumstances and algorithmic models are a crucial basis for research in digital controls and communications.

PREPERATION

In preparation for this lab, various topics from Chapter 3 of the course textbook [1] were acquired. In particular, the following topics come into play in this lab:

1. The Unit Impulse response, $h[n]$
2. System response to external input (i.e the zero-state response).
3. Internal (i.e. Asymptotic) Stability

This included learning the procedures for the following:

- a. Determining impulse response using a closed-form expression.
- b. Finding the unit impulse response by filtering (conceptually & using a computer).
- c. Convolution of two finite-duration signals (how to do using a computer).
- d. Finding other system responses by filtering (conceptually & using a computer).

PART A: Unit Impulse Response

Question 1:

I.
$$y[n] + \frac{1}{(F+1)} y[n-1] - \frac{1}{(F+1)} y[n-2] = \frac{1}{(G+1)} x[n]$$

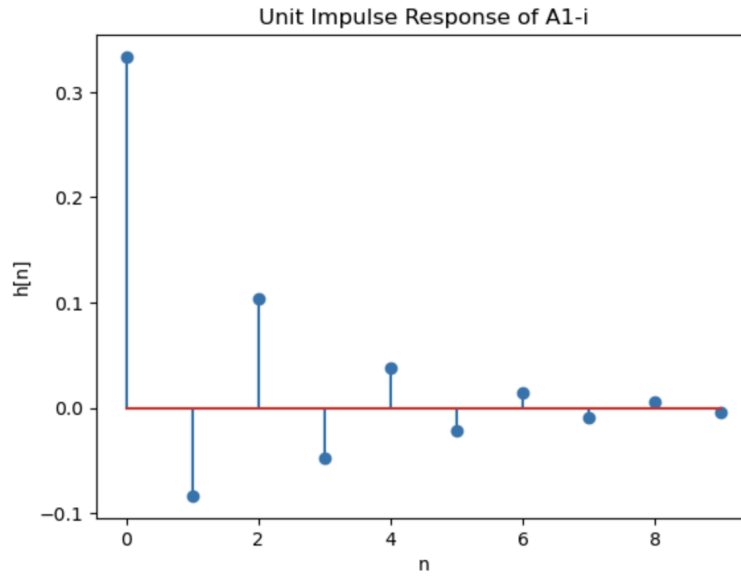


Figure 1: $y[n] + \frac{1}{(4)} y[n-1] - \frac{1}{(4)} y[n-2] = \frac{1}{(3)} x[n]$

II.
$$y[n] + \frac{1}{(H+1)} y[n-2] = x[n]$$

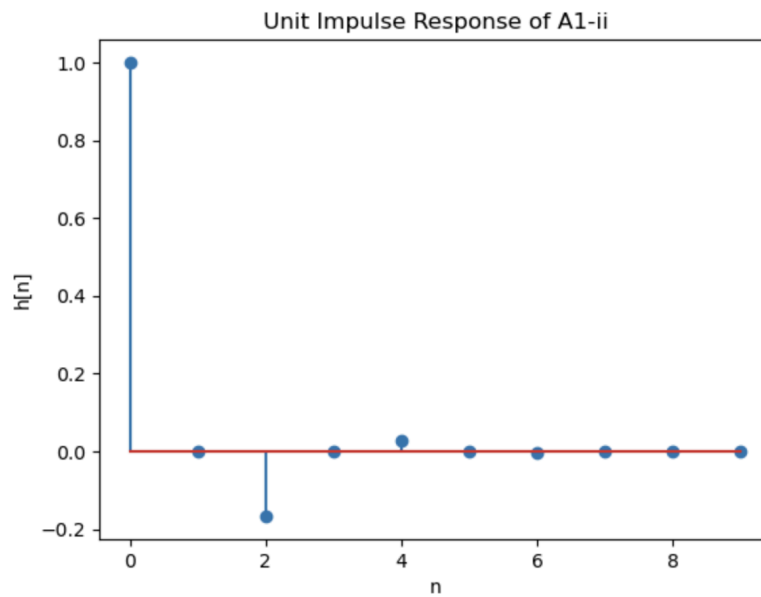


Figure 2: $y[n] + \frac{1}{(6)} y[n-2] = x[n]$

Question 2:

Page:

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Lab 2: Time-Domain Analysis of Discrete-Time Systems

• Part A: Unit Impulse Response

$$2i) y[n] + \frac{1}{4} y[n-1] - \frac{1}{4} y[n-2] = \frac{1}{3} x[n]$$

• Advanced form:

$$y[n+2] + \frac{1}{4} y[n+1] - \frac{1}{4} y[n] = \frac{1}{3} x[n+2]$$

$$\lambda^2 + \frac{1}{4}\lambda - \frac{1}{4} = 0$$

* Using quadratic formula to solve the roots $= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$= \frac{-\frac{1}{4} \pm \sqrt{\frac{1}{16} - 4(1)(-\frac{1}{4})}}{2(1)} = \frac{-\frac{1}{4} \pm \sqrt{\frac{17}{16}}}{2} = -\frac{1}{8} \pm \left(\frac{17}{16}\right)^{\frac{1}{2}} \left(\frac{1}{2}\right)$$

$$= \frac{-1 + \sqrt{17}}{8} \quad \& \quad \frac{-1 - \sqrt{17}}{8}$$

• Characteristic equation:

$$0 = \left[C_1 \left(\frac{-1 + \sqrt{17}}{8} \right)^n + C_2 \left(\frac{-1 - \sqrt{17}}{8} \right)^n \right] u[n]$$

• Particular equation:

$$y_p[n] = A_0 \cdot \delta[n] \quad \text{where } A_0 = \frac{b_n}{a_n}$$

$$= \frac{0}{(-\frac{1}{4})} = 0 \quad \text{Since } b_n = 0 \quad \& \quad a_n = -\frac{1}{4}$$

coeff. of x coeff. of y

$$\therefore y_p[n] = 0$$

②

- Solve for $h[0]$ & $h[1]$ iteratively (using recursion) through the difference equation and substituting $y[n] \rightarrow h[n]$ & $x[n] \rightarrow s[n]$

$$y[n] + \frac{1}{4} y[n-1] - \frac{1}{4} y[n-2] = \frac{1}{3} x[n]$$

$$y[n] = -\frac{1}{4} y[n-1] + \frac{1}{4} y[n-2] + \frac{1}{3} x[n]$$

$$h[n] = -\frac{1}{4} h[n-1] + \frac{1}{4} h[n-2] + \frac{1}{3} s[n]$$

- Put $n=0$

$$\begin{aligned} h[0] &= -\frac{1}{4} h[-1] + \frac{1}{4} h[-2] + \frac{1}{3} s[0] \\ &= 0 + 0 + \frac{1}{3} \Rightarrow h[0] = \frac{1}{3} \end{aligned}$$

- Put $n=1$

$$\begin{aligned} h[1] &= -\frac{1}{4} h[0] + \frac{1}{4} h[-1] + \frac{1}{3} s[1] \\ &= \left(-\frac{1}{4}\right)\left(\frac{1}{3}\right) + 0 + 0 \Rightarrow h[1] = -\frac{1}{12} \end{aligned}$$

- Solve for C_1 & C_2 using initial conditions of $h[0] = \frac{1}{3}$ & $h[1] = -\frac{1}{12}$

$$h[n] = \left[C_1 \left(\frac{-1 + \sqrt{17}}{8} \right)^n + C_2 \left(\frac{-1 - \sqrt{17}}{8} \right)^n \right] u[n]$$

$$h[0] \Rightarrow \frac{1}{3} = C_1 + C_2 \quad \& \quad h[1] \Rightarrow -\frac{1}{12} = C_1 \left(\frac{-1 + \sqrt{17}}{8} \right) + C_2 \left(\frac{-1 - \sqrt{17}}{8} \right)$$

$$C_1 = \frac{1}{3} - C_2 \quad \text{and} \quad -\frac{1}{12} = \left(\frac{1}{3} - C_2 \right) \left(\frac{-1 + \sqrt{17}}{8} \right) + C_2 \left(\frac{-1 - \sqrt{17}}{8} \right)$$

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$$\frac{-1}{12} = \left(\frac{1}{3}\right) \left(\frac{-1+\sqrt{17}}{8}\right) + (-C_2) \left(\frac{-1+\sqrt{17}}{8}\right) + C_2 \left(\frac{-1-\sqrt{17}}{8}\right)$$

$$\frac{-1}{12} = \frac{-1+\sqrt{17}}{24} + \cancel{\frac{C_1}{8}} - \frac{C_2\sqrt{17}}{8} - \cancel{\frac{C_2}{8}} - \frac{C_2\sqrt{17}}{8}$$

$$\frac{-1+\sqrt{17}}{24} = -\frac{2C_2\sqrt{17}}{8} \Rightarrow C_2 = \frac{-1+\sqrt{17}}{3} \left(\frac{-1}{2\sqrt{17}}\right) = \frac{1+\sqrt{17}}{6\sqrt{17}}$$

$$\boxed{C_2 = \frac{1+\sqrt{17}}{6\sqrt{17}}} \quad \& \quad C_1 = \frac{1}{3} - \left(\frac{1+\sqrt{17}}{6\sqrt{17}}\right) = \boxed{\frac{\sqrt{17}-1}{6\sqrt{17}}} = C_1$$

$$\therefore h[n] = \left[\left(\frac{\sqrt{17}-1}{6\sqrt{17}}\right) \left(\frac{-1+\sqrt{17}}{8}\right)^n + \left(\frac{1+\sqrt{17}}{6\sqrt{17}}\right) \left(\frac{-1-\sqrt{17}}{8}\right)^n \right] u[n]$$

where $h[n]$ is the unit impulse response and the characteristic modes are given as follows:

$$\left(\frac{-1+\sqrt{17}}{8}\right)^n \quad \& \quad \left(\frac{-1-\sqrt{17}}{8}\right)^n$$

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$$211) y[n] + \frac{1}{6} y[n-2] = x[n]$$

• Advanced form:

$$y[n+2] + \frac{1}{6} y[n] = x[n+2]$$

• Substitute operator notation:

$$z^2 + \frac{1}{6} = 0$$

* Using quadratic formula to solve the roots = $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$= \frac{0 \pm \sqrt{0 - (4)(1)(\frac{1}{6})}}{2(1)} = \frac{\pm \sqrt{-\frac{2}{3}}}{2}$$

$$= j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \quad \& \quad -j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right)$$

• Characteristic equation:

$$0 = \left[C_1 \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right) + C_2 \left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right) \right] u[n]$$

• Particular equation:

$$y_p[n] = A_0 \cdot \delta[n] \quad \text{where } A_0 = \frac{b_n}{a_n}$$

$$= \frac{0}{\frac{1}{6}} = 0 \quad \text{Since } \underbrace{b_n = 0}_{\text{coeff. of } x} \quad \& \quad \underbrace{a_n = \frac{1}{6}}_{\text{coeff. of } y}$$

$$\therefore y_p[n] = 0$$

5

- Solve for $h[0]$ & $h[1]$ iteratively (using recursion) through the difference equation and substituting $y[n] \rightarrow h[n]$ & $x[n] \rightarrow \delta[n]$

$$y[n] + \frac{1}{6}y[n-2] = x[n]$$

$$y[n] = -\frac{1}{6}y[n-2] + x[n]$$

$$h[n] = -\frac{1}{6}h[n-2] + \delta[n]$$

- Put $n=0$

$$\begin{aligned} h[0] &= -\frac{1}{6}h[-2] + \delta[0] \\ &= 0 + 1 \Rightarrow h[0] = 1 \end{aligned}$$

- Put $n=1$

$$\begin{aligned} h[1] &= -\frac{1}{6}h[-1] + \delta[1] \\ &= 0 + 0 \Rightarrow h[1] = 0 \end{aligned}$$

- Solve for C_1 & C_2 using initial conditions of $h[0]=1$ & $h[1]=0$

$$h[n] = \left[C_1 \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^n + C_2 \left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^n \right] u[n]$$

$$h[0] \Rightarrow 1 = C_1 + C_2 \quad \& \quad h[1] \Rightarrow 0 = C_1 \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right) + C_2 \left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)$$

$$\begin{aligned} C_1 &= 1 - C_2 \quad \& \quad C_2 = \frac{-C_1 \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)}{\left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)} = -C_1(-1) = C_1 \end{aligned}$$

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$$C_1 = 1 - \left(\frac{-C_1 \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)}{\left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)} \right)$$

$$C_1 = 1 + \frac{C_1 \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)}{\left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)}$$

$$\frac{C_1 - C_1 \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)}{\left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)} = 1$$

$$C_1 \left(1 - \frac{\left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)}{\left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)} \right) = 1$$

$$C_1 = \frac{1}{1 - \frac{\left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)}{\left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)}} = \frac{1}{1 - (-1)} = \frac{1}{1 - (-1)} = \frac{1}{2} = C_1$$

$$C_1 = \frac{1}{2} = 1 - C_2 \Rightarrow C_2 = \frac{1}{2}$$

$$C_1 = C_2 = \frac{1}{2}$$

$$\therefore h[n] = \left[\left(\frac{1}{2} \right) \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^n + \left(\frac{1}{2} \right) \left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^n \right] u[n]$$

where $h[n]$ is the unit impulse response and the characteristic modes are given as follows: $\left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^n$ & $\left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^n$

Question 3:

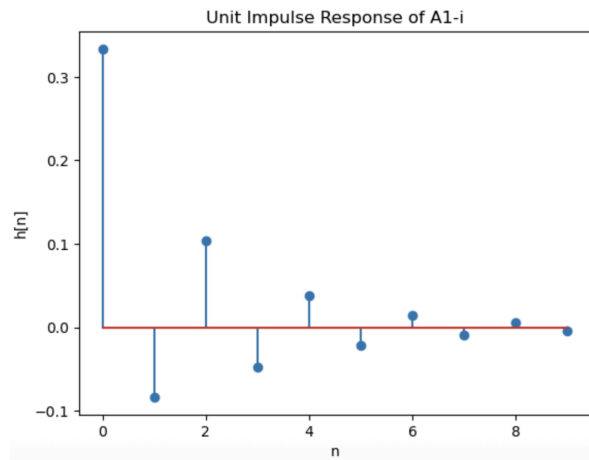


Figure 1: $h[3] = -\frac{3}{64} = -0.04688$

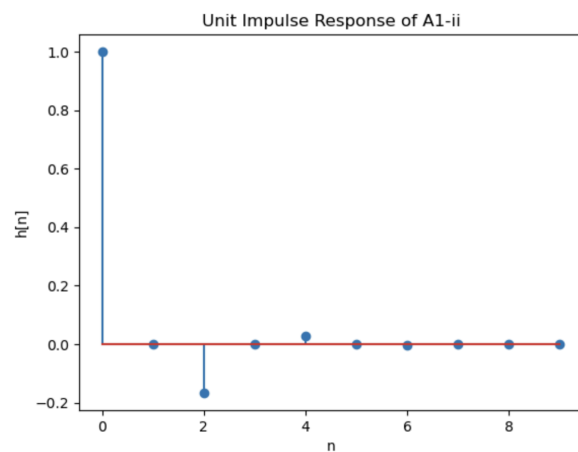


Figure 2: $h[3] = 0$

As per the calculated values for $h[3]$ in parts Ai and Aii, the corresponding graphs (Figure 1 and Figure 2) align with the theoretical calculations. This proves that the hand-written calculations were performed correctly and the Python script is identical in theory.

(7)

3i) $h[n] = \left[\left(\frac{\sqrt{17}-1}{6\sqrt{17}} \right) \left(\frac{-1+\sqrt{17}}{8} \right)^n + \left(\frac{1+\sqrt{17}}{6\sqrt{17}} \right) \left(\frac{-1-\sqrt{17}}{8} \right)^n \right] u[n]$

$h[3] = \left(\frac{\sqrt{17}-1}{6\sqrt{17}} \right) \left(\frac{-1+\sqrt{17}}{8} \right)^3 + \left(\frac{1+\sqrt{17}}{6\sqrt{17}} \right) \left(\frac{-1-\sqrt{17}}{8} \right)^3 \quad (1)$

$= \frac{\sqrt{17}-1}{6\sqrt{17}} \left(\frac{-1+\sqrt{17}}{8} \right) \left(\frac{-1+\sqrt{17}}{8} \right) \left(\frac{-1+\sqrt{17}}{8} \right) + \frac{1+\sqrt{17}}{6\sqrt{17}} \left(\frac{-1-\sqrt{17}}{8} \right) \left(\frac{-1-\sqrt{17}}{8} \right) \left(\frac{-1-\sqrt{17}}{8} \right)$

$= \frac{\sqrt{17}-1}{6\sqrt{17}} \left(\frac{2\sqrt{17}-52}{512} \right) + \frac{1+\sqrt{17}}{6\sqrt{17}} \left(\frac{-2\sqrt{17}-52}{512} \right)$

$= \frac{392-72\sqrt{17}}{3072\sqrt{17}} + \frac{1+\sqrt{17}}{6\sqrt{17}} \left(\frac{-52-20\sqrt{17}}{512} \right)$

$= \frac{392-72\sqrt{17}}{3072\sqrt{17}} + \frac{-392-72\sqrt{17}}{3072\sqrt{17}} = \frac{-144\sqrt{17}}{3072\sqrt{17}} = \frac{-144}{3072} = \frac{-9}{192} = \frac{-3}{64}$

$h[3] = -\frac{3}{64}$

3ii) $h[n] = \left[\left(\frac{1}{2} \right) \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^n + \left(\frac{1}{2} \right) \left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^n \right] u[n]$

$h[3] = \left[\left(\frac{1}{2} \right) \left(j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^3 + \left(\frac{1}{2} \right) \left(-j \left(\frac{2}{3} \right)^{\frac{1}{2}} \left(\frac{1}{2} \right) \right)^3 \right] \quad (1)$

$= \frac{1}{2} \left(-j \left(\frac{2}{3} \right)^{\frac{3}{2}} \left(\frac{1}{8} \right) \right) + \frac{1}{2} \left(j \left(\frac{2}{3} \right)^{\frac{3}{2}} \left(\frac{1}{8} \right) \right)$

$h[3] = 0$

PART B: Zero-Input Response

Question 1:

I. $y[n] - \frac{2(D+1)}{10} y[n-1] - \frac{1}{10} y[n-2] = 2x[n]$

$$y[n] - \frac{2}{5} y[n-1] - \frac{1}{10} y[n-2] = 2x[n] \quad [2]$$

- **Zero-Input Response**

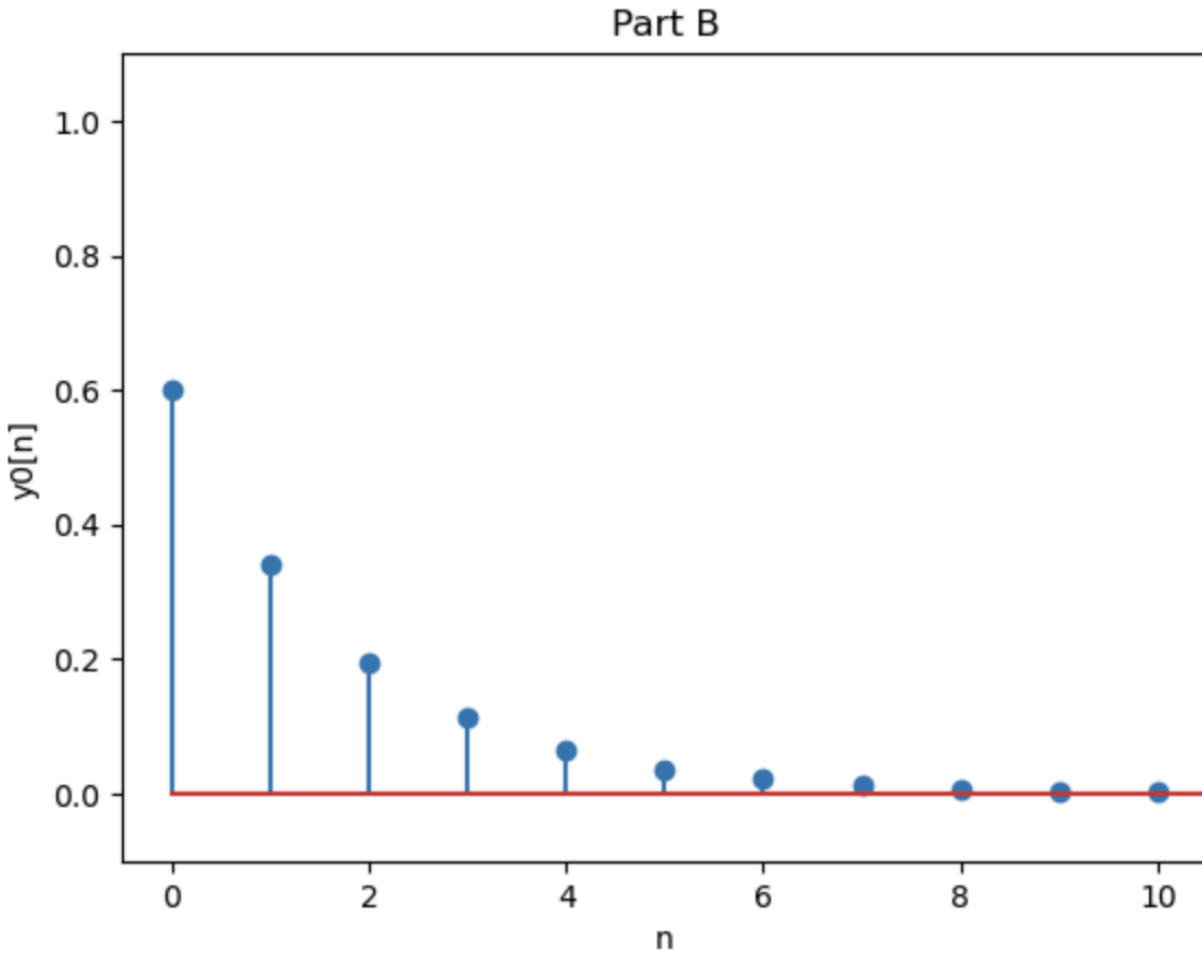


Figure 3: $y[n] - \frac{2}{5} y[n-1] - \frac{1}{10} y[n-2] = 2x[n]$

The above system had its Zero-Input response plotted using Python. It also has two initial conditions, which are given as $y[-1] = 1$ & $y[-2] = 2$.

PART C: Zero-State Response

Question 1:

The system introduced in Part B is calculated and plotted as per its zero-state response, given the following input:

$$x[n] = 2\cos\left(\frac{2\pi n}{(D+1)}\right)(u[n] - u[n - (D + 1)]) \quad [3]$$

- **Zero-State Response**

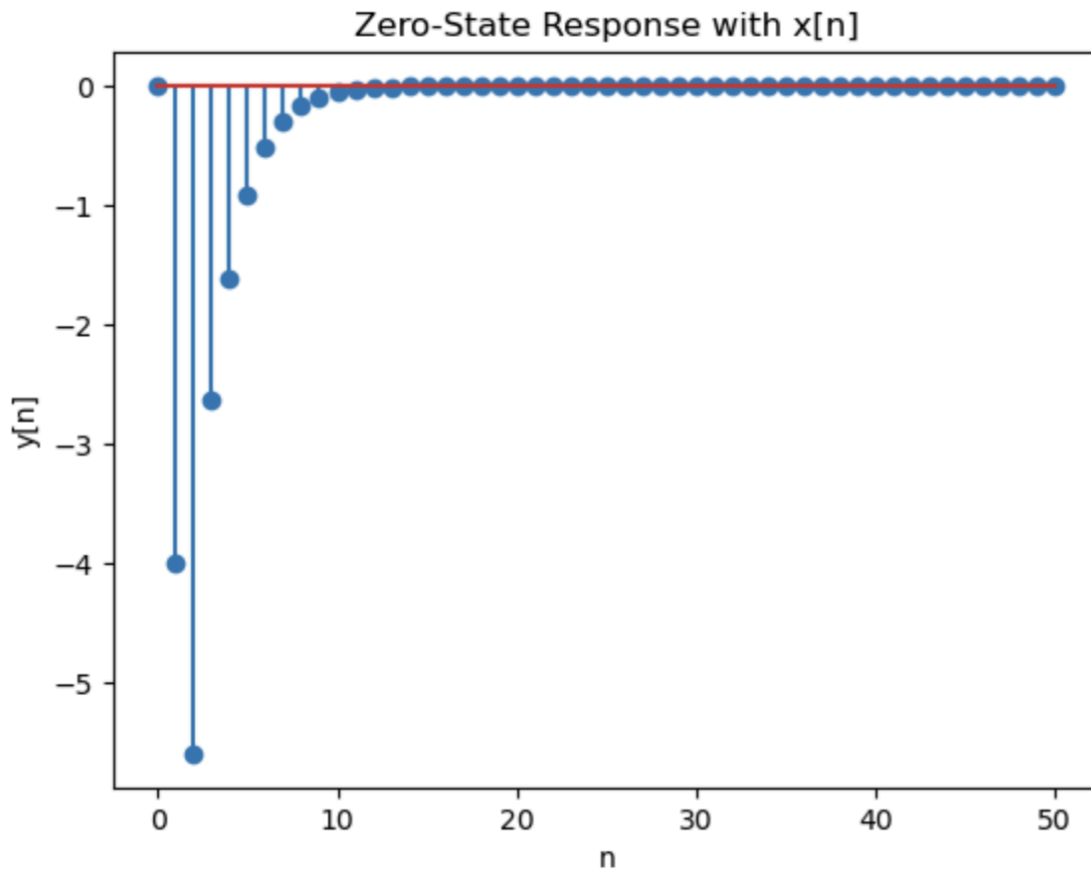


Figure 4: $x[n] = 2\cos(\pi n)(u[n] - u[n - 2])$

PART D: Total Response

Question 1:

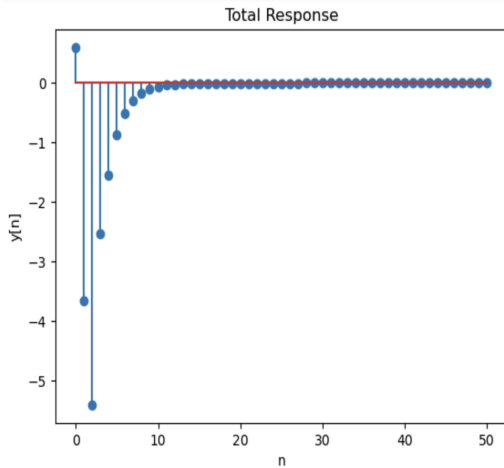


Figure 5: Total response using lfilter

Question 2:

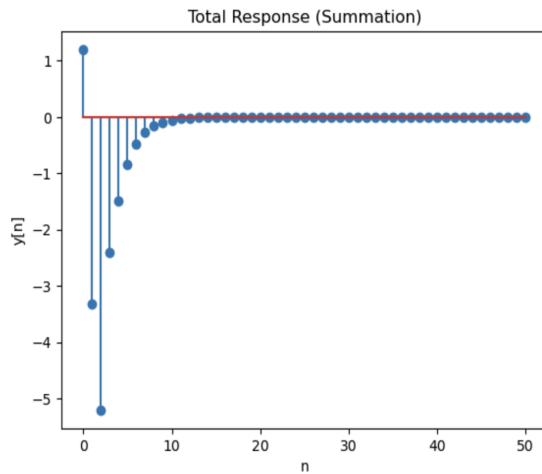


Figure 6: Total Response (adding B and C)

- **Proof (Question 2):**

The total response is given by initial conditions $y[-1] = 1$ and $y[-2] = 2$.

$$y[n] - \frac{2}{5} y[n - 1] - \frac{1}{10} y[n - 2] = 2x[n] \quad [2]$$

- **Part B (Zero-Input Response)**

$$x[n] = 2\cos(\pi n)(u[n] - u[n - 2]) \quad [3]$$

- **Part C (Zero-State Response)**

By substituting equation [3] into equation [2], the resulting waveform:

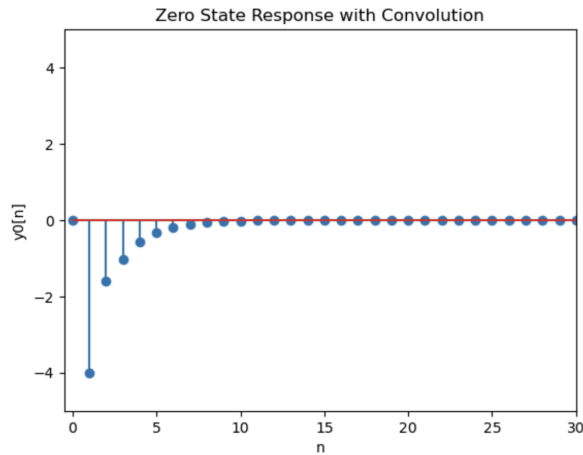
$$y[n] - \frac{2}{5} y[n - 1] - \frac{1}{10} y[n - 2] = 2[2\cos(\pi n)(u[n] - u[n - 2])]$$

$$y[n] - \frac{2}{5} y[n - 1] - \frac{1}{10} y[n - 2] = 4\cos(\pi n)(2u[n] - 2u[n - 2])$$

Therefore, this is feasible based on the graphs displayed. The Total Response has been double-checked by doing both calculations. Adding and plotting them on the same graph to check and make sure that both graphs would be identical, and they are as provided in Figures 5 & Figure 6.

PART E: Convolution and System Stability

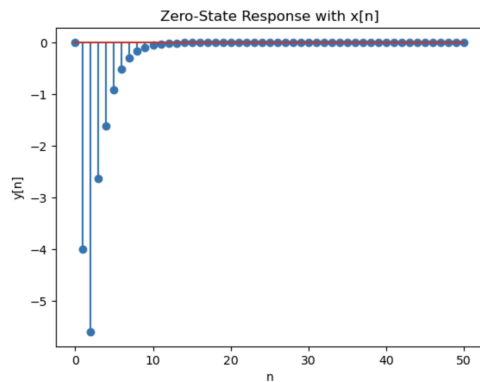
Question 1:



The SciPy Python library's convolve command was used to compute the zero-state response of the system introduced in part B.

Figure 7: Zero-state response found using convolution

Question 2:



The results of the zero-state response using convolution (Figure 7) are identical to that of part C using the filter method (Figure 4).

Figure 4: $x[n] = 2\cos(\pi n)(u[n] - u[n - 2])$

Question 3:

Yes, the system is asymptotically stable. Asymptotic stability in a discrete-time LTI (Linear Time-Invariant) system means that the zero-input response (or the impulse response) decays to zero as time, n, approaches infinity.

Using the coefficients provided, $Q = 1$, $-\frac{2}{5}$, $-\frac{1}{10}$, we can find the poles by solving

for the roots of the characteristic equation: $z^2 - \frac{2}{5}z - \frac{1}{10} = 0$, using the quadratic formula, $z_1 = 0.574$ and $z_2 = -0.174$.

Therefore, since the absolute value of both poles is less than 1 ($|z_1| < 1$ and $|z_2| < 1$), they both reside within the unit circle, mathematically proving the system is asymptotically stable.

PART E: Moving Average Filter

Question 1:

The constant coefficient difference equation for the filter is given by $y[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[n - k]$, for its impulse response $h[n] = \frac{1}{N} \sum_{k=0}^{N-1} \delta[n - k]$.

Question 2:

```
def movingAvg(N):
    # System is FIR, Therefore a = 1
    a = [1] # FeedBack coefficients

    # all coefficients are 1/N
    b = np.ones(N)/N # Feed forward coefficients
    return b, a

def CoeffCalc(N):
    output = [[], [1]]
    for i in range(0, N, 1):
        output[0].append(1/N)
    return output

hAtn = lambda N: signal.filter(CoeffCalc(N)[0], CoeffCalc(N)[1], delta(n))

print("\nThe constant coefficient impulse response for N=4 is given by h[n]=",
      CoeffCalc(4)[0])

print("\nThe constant coefficient impulse response for N=8 is given by h[n]=",
      CoeffCalc(8)[0])

print("\nThe constant coefficient impulse response for N=12 is given by
      h[n]=", CoeffCalc(12)[0])
```

Function 1: Computes the Parameters necessary to implement an N-point moving-average filter using SciPy's lfilter command

The above code was used to calculate the parameters necessary to implement the N-point moving average filter in vectors b and a, given a scalar input N. From the following output of the code, the constant coefficients are found for the N values of 4, 8, and 12:

The constant coefficient impulse response for N=4 is given by $h[n] = [0.25, 0.25, 0.25, 0.25]$

The constant coefficient impulse response for N=8 is given by $h[n] = [0.125, 0.125, 0.125, 0.125, 0.125, 0.125, 0.125, 0.125]$

The constant coefficient impulse response for N=12 is given by $h[n] = [0.0833333333333333, 0.0833333333333333, 0.0833333333333333, 0.0833333333333333, 0.0833333333333333, 0.0833333333333333, 0.0833333333333333, 0.0833333333333333, 0.0833333333333333, 0.0833333333333333, 0.0833333333333333, 0.0833333333333333]$

Question 3:

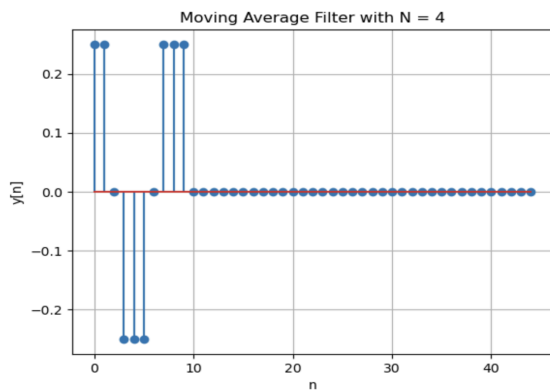


Figure 8: $x[n]$ for a 4-point averaging filter

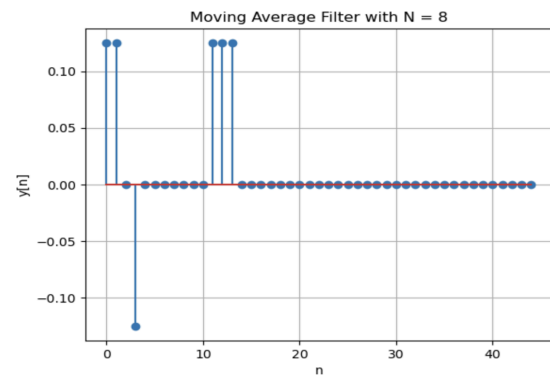


Figure 9: $x[n]$ for a 8-point averaging filter

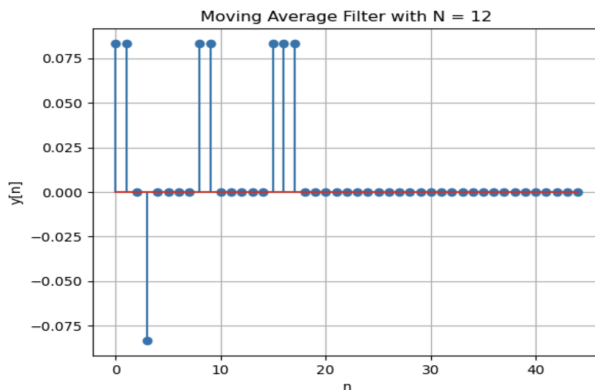


Figure 10: $x[n]$ for a 12-point averaging filter

The input for the N-point filter is given by:

$$x[n] = \cos\left(\frac{\pi n}{2}\right) - \delta[n-3] + \delta[n-6]$$

The behaviour of this filter takes the average of the N points around it, lowering the amplitude of the cosine plot to a higher degree as N increases. This is because with a greater N, more points (including those with opposite polarities) are taken in the average, further pushing the average down. This reduction in magnitude is clearly illustrated in Figures 8 through 10.

REFERENCES

[1] Lathi, B. P., & Green, R. A. (2018). Linear Systems and Signals (3rd ed.). Oxford University Press.

APPENDIX

PART A

Question 1-i:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import signal

n = np.arange(0,10)
delta = lambda n:(1.0*(n==0))
E = np.array([1/3, 0, 0])
Q = np.array([1, 1/4, -1/4])

#signal.lfilter(Q, E, delta(n))
h = signal.lfilter(E, Q, delta(n))

plt.stem(n,h)
plt.title('Unit Impulse Response of A1-i')
plt.xlabel('n')
plt.ylabel('h[n]')
plt.show()
```

Question 1-ii:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import signal

n = np.arange(0,10)
delta = lambda n:(1.0*(n==0))
E = np.array([1, 0, 0])
Q = np.array([1, 0, 1/6])

#signal.lfilter(Q, E, delta(n))
h = signal.lfilter(E, Q, delta(n))

plt.stem(n,h)
plt.title('Unit Impulse Response of A1-ii')
plt.xlabel('n')
plt.ylabel('h[n]')
plt.show()
```

PART B

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import lfilter, lfiltic

b = [2, 0, 0]
a = [1, -2/5, -1/10]

z_i = lfiltic(b, a, [1, 2])
n = np.arange(0, 51)
x = np.zeros(n.shape)
y_0, _ = lfilter(b, a, x, zi=z_i)

plt.stem(n, y_0)
plt.title('Part B')
plt.axis([-0.5, 10.5, -0.1, 1.1])
plt.xlabel('n')
plt.ylabel('y0[n]')
plt.show()
```

PART C

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import lfilter, lfiltic, square

E = [2, 0, 0]
Q = [1, -2/5, -1/10]

n = np.arange(0, 51)
x = 2 * np.cos(np.pi * n) * (np.heaviside(n, 0) - np.heaviside(n - 2, 2)) # where  $D = 1 + 1 = 2$ 
y = lfilter(E, Q, x)

plt.stem(n, y)
plt.title('Zero-State Response with x[n]')
plt.xlabel('n')
plt.ylabel('y[n]')
plt.show()
```

PART D

Question 1:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import lfilter, lfiltic, square

b = [2, 0, 0]
a = [1, -2/5, -1/10]

n = np.arange(0, 51)
x = 2 * np.cos(np.pi * n) * (np.heaviside(n, 0) - np.heaviside(n - 2, 2))

z_i = lfiltic(b, a, [1, 2])
y, _ = lfilter(b, a, x, zi=z_i)

plt.stem(n, y)
plt.title('Total Response')
plt.xlabel('n')
plt.ylabel('y[n]')
plt.show()
```

Question 2:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import lfilter, lfiltic, square

b = [2, 0, 0]
a = [1, -2/5, -1/10]

n = np.arange(0, 51)
x = 2 * np.cos(np.pi * n) * (np.heaviside(n, 0) - np.heaviside(n - 2, 2))

z_i = lfiltic(b, a, [1, 2])
y1, _ = lfilter(b, a, x, zi=z_i)

z_i = lfiltic(b, a, [1, 2])
y0 = lfilter(b, a, np.zeros_like(n), zi=z_i)[0]

y = y0 + y1

plt.stem(n, y)
plt.title('Total Response (Summation)')
plt.xlabel('n')
plt.ylabel('y[n]')
plt.show()
```

PART E

Question 1:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import lfilter, convolve

n = np.arange(0, 51)
u = np.where(np.mod(n,1)==1, 0.0, 0.0) * np.where(n>=0, 0.0, 0.0)
u2 = np.where(np.mod(n,1)==1, 0.0, 1.0) * np.where(n>=2, 0.0, 1.0)
b = np.array([2, 0, 0])
a = np.array([1, -2/5, -1/10])
x = 2. * np.cos((np.pi * n) / 1) * (u - u2)

delta = np.where(n==0, 0, 1)

h = lfilter(b, a, delta)

y = convolve(x, h)

n = np.arange(0, 101)

plt.stem(n, y)
plt.title('Zero State Response with Convolution')
plt.axis([-0.5, 30, -5, 5])
plt.xlabel('n')
plt.ylabel('y0[n]')
plt.show()
```

PART F

Question 1:

```
def movingAvg(N):
    # System is FIR, Therefore a = 1
    a = [1] # FeedBack coefficients

    # all coefficients are 1/N
    b = np.ones(N)/N # Feed forward coefficients
    return b, a
```

Question 2:

```
def CoeffCalc (N):
    output=[[],[1]]
    for i in range (0,N,1):
        output[0].append(1/N)
    return output

hAtn = lambda N: signal.filter(CoeffCalc(N)[0], CoeffCalc(N)[1], delta(n))
print("\nThe constant coefficient impulse response for N=4 is given by h[n]=", CoeffCalc(4)[0])
print("\nThe constant coefficient impulse response for N=8 is given by h[n]=", CoeffCalc(8)[0])
print("\nThe constant coefficient impulse response for N=12 is given by h[n]=", CoeffCalc(12)[0])
```

Question 3:

```
dfunc = lambda n: 1.0 * (n == 0)
n = np.arange(0, 45)

x = np.cos(np.pi * n / 2) - dfunc(n - 3) + dfunc(n - 6)

N_values = [4, 8, 12]

for N in N_values:
    b, a = movingAvg(N)
    y = signal.lfilter(b, a, x)
    plt.stem(n, y)
    plt.xlabel("n")
    plt.ylabel("y[n]")
    plt.title(f"Moving Average Filter with N = {N}")
    plt.grid()
    plt.show()
```